MVAGENT: MULTI-AGENT VIDEO GENERATION VIA CONSISTENT CONDITION CONSTRUCTION AND SHOT-LEVEL POLICY OPTIMIZATION

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arXiv:2609.30609v1 [cs.CV] 24 Sep 2026

ABSTRACT

Multi-shot agentic video generation requires consistent character appearance, stable spatial layout across camera angles, and continuous character state between shots. When every shot is a separate request to a frozen generator, repeated text does not determine appearance, layout or state. We therefore recast the problem as condition construction and present MVAgent, a multi-agent pipeline whose agents collaborate through typed conditioning inputs. Because an environment image shows one viewpoint, a Spatial Grounding agent samples views from generated camera-traversal clips and anchors each shot to the view matching its framing. As generated shots drift from the plan, an Observer records how each shot ends in a continuity memory, from which a Transition agent builds character action and spatial references for the next shot. An Orchestrator composes these inputs into each request. Since a request reveals its effect only after rendering, we train it by agentic reinforcement learning with Trunk-GDPO, which compares rendered candidates at every shot rather than once per video and continues the best as the trunk. With generator and judges frozen, MVAgent attains the highest cross-shot consistency and narrative-planning quality among the compared methods on ViMax-Bench and is preferred over the strongest agentic baseline in human evaluation.

***Index Terms*—** Agentic video generation, multi-agent collaboration, reinforcement learning

1. INTRODUCTION

Text-to-video models [1] generate high-quality individual shots, but a short film of eight to sixteen shots across several scenes additionally requires consistent character appearance, stable spatial layout and continuous character state across shots. With a black-box generator, separate requests share no internal state, so a recurring character may change appearance, the fixed elements of an environment may shift between viewpoints, and consecutive shots may show incompatible character states. Repeated textual descriptions constrain semantic content but not the visual appearance or spatial arrangement the generator produces.

Existing approaches improve consistency through identity adapters [2], reference attention [3], camera control [4], explicit scene representations [5] or joint multi-shot generation [6]. All of these require generator parameters or internal features that a black-box generator does not expose. Agentic frameworks instead decompose production into roles and coordinate frozen models around them: LLM-directed pipelines [7] plan scenes and shots with director and screenwriter roles, AniMaker [8] adds clip selection and review agents, ViMax [9] coordinates production roles and renders transition clips, NEWTON [10] trains a planner on trajectory-level rewards, and VISTA [11] and UniVA [12] add multi-critic refinement and plan-and-act tool use. In these systems the roles exchange only text plans, prompts and keyframes, which underdetermine what the generator renders, and the roles are fixed prompts that never learn from the shots they produce.

We instead treat condition construction as the interface between narrative planning and frozen video generation, and realise it in MVAgent, a multi-agent pipeline whose agents collaborate through typed conditioning inputs rather than text alone (Fig. 1a), like the structured outputs of MetaGPT [13]. Because a single environment image constrains only the viewpoint it shows, *generative spatial anchoring* (Sec. 2.2) lets a Spatial Grounding agent sample views from camera-traversal clips rendered by the generator and index them by visible fixed elements, so that a view matching the target camera guides each shot. As generated clips depart from the planned states, an Observer stores VLM descriptions of each clip in a *continuity memory* (Sec. 2.3), from which a Transition agent derives character action and spatial references, so that each shot opens from the state actually reached. An Orchestrator composes these inputs into each request. Since a request reveals its effect only after rendering, we train it by agentic reinforcement learning [14], the optimization of LLM agents through multi-step interaction with an environment, here with one costly render per step and one evaluation per video. Instead of crediting every step with that evaluation [15, 10] or relying on states that recur across rollouts [16], *Trunk-GDPO* (Sec. 2.4) branches candidates from each shot state, compares them with group-relative, reward-decoupled advantages [17, 18] and continues the best as the trunk, with asynchronous updates [19] hiding rendering latency. Our contributions are twofold. First, we design a multi-agent collaboration for agentic video generation that recasts cross-shot consistency as condition construction through generative spatial anchoring and observation-grounded continuity. Second, to train Orchestrator, we propose Trunk-GDPO, which supervises it at shot level within a frozen pipeline and improves consistency and human preference over episode-level training.

2. METHOD

Framework overview. MVAgent (Fig. 1a) is a multi-agent pipeline whose agents collaborate through typed conditioning inputs. The Screenwriter fixes the persistent assets and the shot plan, the Spatial Grounding agent supplies a view of the environment for each shot, the Observer and the Transition agent supply the continuity memory and a transition reference, and the Orchestrator composes these inputs into one request for the frozen generator. All roles are frozen ex-

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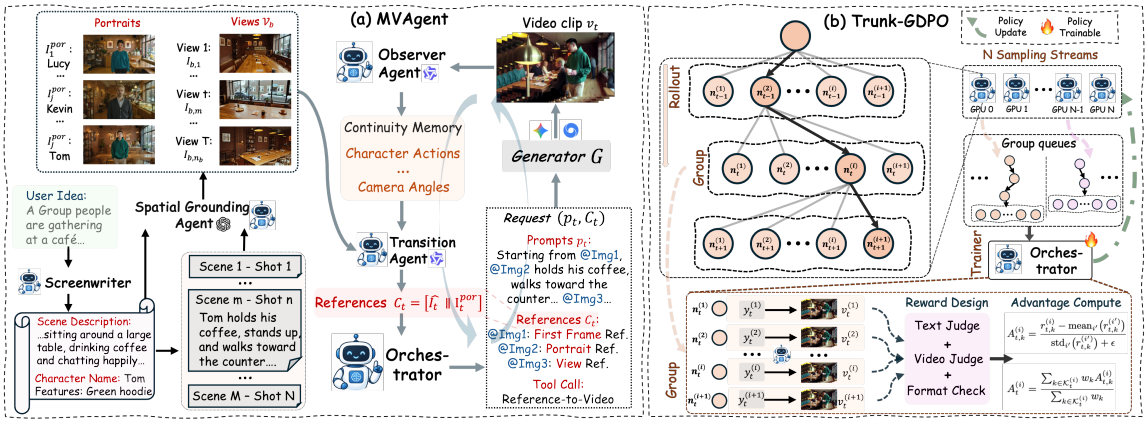


Fig. 1. MVAgent. (a) The Screenwriter and the Spatial Grounding agent supply the persistent assets, the environment image I_b and the multi-view library, which anchor character appearance and spatial layout. The Observer and the Transition agent supply the continuity memory and the reference $\hat{I}_t$, which carry character state. The Orchestrator composes them into each request to the frozen video generator. (b) Trunk-GDPO trains the Orchestrator. A shot-level candidate group shares one state and policy version, all candidates are rendered and judged, components are normalised separately before aggregation, and the trunk advances the video, with asynchronous sampling and training.

cept the Orchestrator's shot-generation policy, which Trunk-GDPO trains with feedback from frozen judges.

2.1. Problem statement

A video consists of T shots produced by a *black-box* video generator G that is accessed only through its conditioning inputs. Shot t receives a prompt p_t and an ordered tuple of reference images $\mathcal{C}_t$, written $[\cdot]$, whose positions carry fixed roles, the first image anchoring the transition into the shot and the remaining images identifying the characters, and yields $v_t \sim G(\cdot \mid p_t, \mathcal{C}_t)$. The Screenwriter provides the persistent assets, namely a portrait I_j^{por} of each character and the fixed elements $\mathcal{E}_b$ of each environment b, and a shot plan that specifies for each shot the cast $\mathcal{P}_t$, the environment b_t, a camera specification κ_t (shot size, camera position and facing), the action and dialogue, and the intended opening and end states. Character appearance and spatial layout, namely the arrangement of $\mathcal{E}_b$ under changes of viewpoint, are addressed by the spatial branch (Sec. 2.2), character state across connected shots by the continuity branch (Sec. 2.3), and adherence to the shot plan by the Orchestrator's request composition (Sec. 2.4). Since G is fixed and text constrains semantics rather than appearance, the problem is to construct requests $\{(p_t, \mathcal{C}_t)\}_{t=1}^{T}$ that carry the persistent assets together with observations of the generated state.

2.2. Generative spatial anchoring

Textual environment descriptions underdetermine spatial layout, so separate requests place recurring elements differently, and a single reference image constrains only the viewpoint it shows. The Spatial Grounding agent therefore anchors each environment b to a multi-view library that is generated rather than reconstructed. From a character-free environment image I_b rendered for $\mathcal{E}_b$, it renders camera-traversal clips $v_b^{\mathrm{trav}} \sim G(\cdot \mid p_b^{\mathrm{trav}}, [I_b])$, where the prompt p_b^{trav} moves the camera through the environment, so that the temporal coherence of the generator keeps one layout in view while the viewpoint changes. Frames sampled from these clips form the candidate set $\mathcal{F}_b$, a candidate joins the view set $\mathcal{U}_b$ only when it is distinct from every view admitted before it in sampling order, and the library $\mathcal{V}_b$ pairs each admitted view with a structured description of its fixed elements and their arrangement, produced by a VLM Φ:

$$\begin{aligned} \mathcal{U}_b &= \{I_b\} \cup \big\{ f \in \mathcal{F}_b : \min\nolimits_{u \in \mathcal{U}_b^{\prec f}} \Delta(f, u) \geq \delta \big\}, \\ \mathcal{V}_b &= \{(I_{b,m}, c_{b,m})\}_{m=0}^{n_b}, \qquad c_{b,m} = \Phi(I_{b,m}), \end{aligned} \tag{1}$$

where $\mathcal{U}_b^{\prec f}$ denotes the views admitted before f, Δ is a pixel-space distance with distinctness threshold δ, and $I_{b,0} = I_b, \ldots, I_{b,n_b}$ enumerate $\mathcal{U}_b$.

At generation time the library is queried with the camera specification: an LLM Λ compares κ_t with the descriptions and returns the view whose fixed elements should be visible under the intended framing, $m_t^\star = \Lambda(\kappa_t, \{c_{b_t,m}\}_m)$, with the environment image as default when no view applies. The retrieved view and its description condition the transition of Sec. 2.3 as a visual reference and a textual layout constraint, which anchor the arrangement of fixed elements while leaving framing to the shot plan.

2.3. Continuity memory

After each shot the Observer, a VLM, describes the camera configuration and the position, orientation and action of each visible character at the end of v_{t-1}, and the description is stored in the continuity memory with the shot plan entry and the request. The Transition agent turns this memory into a reference for character actions and positions by rendering a two-shot transition whose first shot restates the observed state and whose second shot states opening of shot t:

$$\begin{aligned} \hat{I}_t &= \mathrm{Ref}\Big(G\big(\cdot \mid p_t^{\mathrm{2s}}, [f_{t-1}] \,\|\, \mathcal{I}_t^{\mathrm{por}} \,\|\, [I_t^{\mathrm{sp}}]\big)\Big), \\ \mathcal{C}_t &= [\hat{I}_t] \,\|\, \mathcal{I}_t^{\mathrm{por}}, \end{aligned} \tag{2}$$

where $\|$ denotes ordered concatenation, p_t^{2s} is the prompt of this transition, f_{t-1} is the final frame of v_{t-1}, $\mathcal{I}_t^{\mathrm{por}} = [I_j^{\mathrm{por}}]_{j \in \mathcal{P}_t}$ holds the portraits of the cast, the spatial reference I_t^{sp} is the selected view $I_{b_t,m_t^\star}$ when the environment is unchanged and the environment image I_{b_t} otherwise, and Ref extracts the first frame after the cut. The Spatial Grounding agent verifies that $\hat{I}_t$ preserves the arrangement of fixed elements described by $c_{b_t,m_t^\star}$, irrespective of illumination, and requests one regeneration with a new seed when the verification fails. The selected continuation updates the memory for the next shot.

2.4. Trunk-GDPO

The Orchestrator's policy selects a generation mode and writes a prompt specifying reference usage, planned actions, camera behavior and applicable continuity constraints. An episode is one video and an agent step is one request. A terminal reward evaluates a completed video only once, whereas each rendered shot can be evaluated locally, so Trunk-GDPO samples K candidates from the same shot state, executes and evaluates all of them, and uses their relative rewards to supervise the policy, while one candidate continues the rollout. This candidate is the *trunk*, by analogy with expanding one branch of a search tree. A video of T shots yields T comparison groups from TK candidate renders, the same number as K complete videos, and supervises individual decisions under a shared context without estimating their delayed effects.

Groups and advantages. The state s_t collects the shot plan entry, the persistent assets, the continuity memory, the selected view $I_{b_t,m_t^\star}$, the reference tuple $\mathcal{C}_t$ and the available generation modes $\mathcal{M}_t$. The policy emits a structured output y_t, a JSON object naming the mode and the prompt, a fixed executor validates it into the request $a_t = \mathrm{Exec}(s_t, y_t) = (p_t, \mathcal{C}_t)$, and the feedback e_t comprises the rendered clip v_t and the judge assessments. With the backends used here reference-to-video is the only mode, so learning affects prompt construction. At each shot, K candidates $\{y_t^{(i)}\}_{i=1}^{K} \sim \pi_{\theta_{\mathrm{old}}}(\cdot \mid s_t)$ share one state and one behavior-policy version and receive component rewards $r_{t,k}^{(i)} = R_k(s_t, y_t^{(i)}, e_t^{(i)})$ for $k \in \mathcal{K}$. Because the components differ in scale and applicability, each available component is normalised across the candidates before aggregation, following GDPO [18]:

$$A_{t,k}^{(i)} = \frac{r_{t,k}^{(i)} - \mathrm{mean}_{i'}\big(r_{t,k}^{(i')}\big)}{\mathrm{std}_{i'}\big(r_{t,k}^{(i')}\big) + \epsilon}, \qquad A_t^{(i)} = \frac{\sum_{k \in \mathcal{K}_t^{(i)}} w_k A_{t,k}^{(i)}}{\sum_{k \in \mathcal{K}_t^{(i)}} w_k}, \tag{3}$$

where $\mathcal{K}_t^{(i)}$ contains the components available for candidate i, tied components yield zero advantage, and no value function is learned. The candidate with the highest combined advantage advances shot-level video generation and updates the memory,

$$s_{t+1} = \mathcal{T}\big(s_t,\, y_t^{(i_t^\star)},\, e_t^{(i_t^\star)}\big), \qquad i_t^\star = \arg\max_i \, A_t^{(i)}. \tag{4}$$

Objective and LoRA policy. Let $\rho_{t,j}^{(i)}$ denote the current-to-behavior probability ratio for token j of candidate i. The policy maximises a clipped surrogate [20] regularised towards the frozen reference policy π_{ref}:

$$\begin{aligned}\mathcal{J}(\theta) = \mathbb{E}\Big[& \frac{1}{\sum_i |y_t^{(i)}|} \sum_{i=1}^{K} \sum_{j=1}^{|y_t^{(i)}|} \min\big\{\rho_{t,j}^{(i)} A_t^{(i)}, \\ & \mathrm{clip}(\rho_{t,j}^{(i)}, 1-\varepsilon_\ell, 1+\varepsilon_h) A_t^{(i)}\big\}\Big] - \beta\, \mathbb{D}_{\mathrm{KL}}\big(\pi_\theta \,\|\, \pi_{\mathrm{ref}}\big),\end{aligned} \tag{5}$$

where the asymmetric clipping follows DAPO [17], response tokens are averaged within the group, each group supports one optimizer update, and the KL penalty uses the per-token k_3 estimator. Training updates low-rank adapters [21] on the attention projections with zero initial effective update, so the initial policy is the base model and disabling the adapters recovers π_{ref}, and only adapter parameters move between the trainer and the sampling workers.

Asynchronous training. To overlap rendering with optimization, N sampling streams generate videos and enqueue each completed group, and one trainer performs one update per group and publishes adapter parameters every B updates (Fig. 1b). Streams reload parameters only at shot boundaries, so the K candidates of a group share one behavior policy while different shots of a video may use different versions. Groups older than $\sigma_{\max}$ versions are discarded, so the policy staleness of every update is bounded, and stored token IDs and behavior log-probabilities let the trainer recompute current-policy probabilities in the same contexts, which keeps the ratios consistent with the stored samples.

Reward components. The components in Eq. (3) are a format, a text and a video reward, $k \in \mathcal{K} = \{\mathrm{fmt}, \mathrm{txt}, \mathrm{vid}\}$, whose advantages are weighted by $w_{\mathrm{fmt}} = 0.15$, $w_{\mathrm{txt}} = 0.35$ and $w_{\mathrm{vid}} = 0.50$. The format reward $r_{t,\mathrm{fmt}}^{(i)}$ is 1 if the executor accepts $y_t^{(i)}$ and 0 otherwise, and weighs least because a larger weight rewarded well-formed but empty prompts. The text reward $r_{t,\mathrm{txt}}^{(i)}$ is one fifth of the text judge's mean 1–5 rating on criteria including plan adherence, visual specificity and reference usage. For the video reward $r_{t,\mathrm{vid}}^{(i)}$, the video judge ranks the K clips on acting (act), physical plausibility (phy) and camera work (cam), which lack a reference, but checks consistency (cons) against the portraits and the selected view, since ranking would still reward the best of a failing group:

$$\begin{aligned} r_{t,\mathrm{vid}}^{(i)} &= \textstyle\sum_{d \in \mathcal{D}} \lambda_d\, r_{t,d}^{(i)}, \qquad \mathcal{D} = \{\mathrm{act}, \mathrm{phy}, \mathrm{cam}, \mathrm{cons}\}, \\ r_{t,d}^{(i)} &= \begin{cases} 1 - \mathrm{rk}_{t,d}^{(i)}/(K-1), & d \in \{\mathrm{act}, \mathrm{phy}, \mathrm{cam}\}, \\ q_{t,+}^{(i)} / \big(q_{t,+}^{(i)} + q_{t,-}^{(i)}\big), & d = \mathrm{cons}, \end{cases} \end{aligned} \tag{6}$$

where λ_d are fixed weights summing to one, $\mathrm{rk}_{t,d}^{(i)}$ is the zero-based rank of clip i, and $q_{t,+}^{(i)}$ and $q_{t,-}^{(i)}$ count the appearance attributes and fixed elements the clip matches and contradicts.

3. EXPERIMENTS

Setup. Each final shot uses one policy output at inference, without best-of-K selection or post-generation repair. Unless a table row names another LLM, the LLM-based agents use GPT-5.4, the VLM-based agents use Gemini 3 Pro and the video generator is Veo 3.1. For Trunk-GDPO training only the LLM in the Orchestrator role is replaced by the open-source Qwen3-32B [22] with rank-64 LoRA ($\alpha = 128$), and Table 4 uses this configuration. The frozen text and video judges are qwen3.8-max and Gemini 3.1 Pro. Training uses $K = 4$, $(\varepsilon_\ell, \varepsilon_h) = (0.2, 0.3)$, $\beta = 10^{-3}$, learning rate 10^{-5}, $B = 4$, $\sigma_{\max} = 5$, $N = 4$ sampling streams on four GPUs, one trainer on four GPUs.

Benchmarks and metrics. ViMax-Bench [9] contains 35 stories (12 Medium with two scenes and 8–10 shots, 23 Long with three to four scenes and 12–16 shots) covering character persistence, background persistence and multi-person interaction. Following its protocol, we encode each shot with ViCLIP [23] and report intra-scene (IC), cross-scene (CC) and global (GC) cosine similarity. Narrative quality follows the rubric of [9] on 50 NarrativeQA novels [24], where GPT-5.4 scores character behavior consistency (CBC), narrative coherence (NC), plot pacing and rhythm (PPR), visual specificity and shot usability (VSU), scene transition quality (STQ) and faithfulness (NF) on a 1–5 scale. In the pairwise protocol of [9], three raters compare video pairs shown in random order on cross-scene consistency (CSC), semantic following (SF) and aesthetic quality (AQ), reported as preference rates in percent.

Cross-shot consistency. Table 1 reports results alongside the baseline scores from [9]. With Gemini 3 Pro and Veo 3.1, MVAgent achieves the highest score in eight of nine columns, including overall GC (0.5689 versus 0.5571 for ViMax), and with Gemini 3.1 Flash and Veo 3.1 it exceeds ViMax in every column, both under the same language and video backends.

Table 1. Cross-shot consistency on ViMax-Bench (ViCLIP cosine similarity for cross-scene CC, intra-scene IC and global GC, higher is better). Agentic rows name the LLM and the generator, bold and underline mark the highest and second-highest scores.

Method	Medium			Long			Overall		
	CC	IC	GC	CC	IC	GC	CC	IC	GC
Veo 3.1	0.5536	0.5640	0.5582	0.4687	0.4778	0.4704	0.4978	0.5074	0.5005
IC-LoRA	0.5398	0.5713	0.5537	0.5015	0.5328	0.5087	0.5147	0.5460	0.5241
StoryDiffusion	0.5606	0.5796	0.5691	0.5007	0.5195	0.5048	0.5212	0.5401	0.5269
HoloCine	0.5366	0.5892	0.5597	0.4940	0.5565	0.5080	0.5098	0.5686	0.5272
ViMax (Gemini 3.1 Flash, Veo 3.1)	0.5681	0.5914	0.5785	0.5088	0.5402	0.5157	0.5291	0.5577	0.5372
ViMax (Gemini 3 Pro, Veo 3.1)	0.5886	**0.6046**	0.5957	0.5298	0.5629	0.5369	0.5500	0.5772	0.5571
MVAgent (Gemini 3.1 Flash, Veo 3.1)	0.5926	0.6012	0.5890	0.5126	0.5494	0.5271	0.5403	0.5602	0.5450
MVAgent (Gemini 3 Pro, Veo 3.1)	**0.5997**	0.6029	**0.5978**	**0.5348**	**0.5704**	**0.5437**	**0.5611**	**0.5863**	**0.5689**

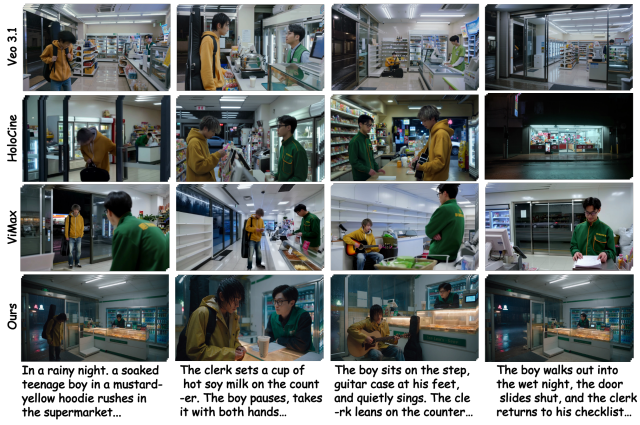


Fig. 2. Qualitative comparison of three conditioning strategies for a multi-shot story. Each row shows keyframes generated with the same video model, story and camera specifications.

Table 2. Narrative-planning quality (1–5 scale) with different frozen agent backbones, following novel-to-video protocol of [9].

Backbone	CBC	NC	PPR	VSU	STQ	NF	Avg.
Ours-Gemini 3 Pro	4.37	4.08	4.19	4.83	4.01	2.64	4.02
Ours-Gemini 3.1 Flash	4.29	3.96	4.20	4.76	4.10	2.57	3.98
Ours-GPT-5.4	**4.54**	**4.22**	**4.41**	**4.90**	**4.23**	**2.82**	**4.19**

Qualitative comparison. Fig. 2 compares three conditioning strategies with the generator, story and camera specifications held fixed. MVAgent preserves character appearance across the four shots, and the store retains its arrangement of doors, counters and shelves under different framings. Text-only conditioning changes the counter, freezer and door arrangement and the customer's appearance despite repeated descriptions, and inconsistent descriptions produce larger changes in both.

Narrative quality and LLM backbone. MVAgent achieves the highest mean score in the novel-to-video comparison (4.256 versus 4.107 for the strongest agentic baseline). Its advantage does not extend to faithfulness, which is consistent with the absence of retrieval over the novel. Table 2 compares frozen agent backbones with the other roles fixed. GPT-5.4 obtains the highest score on all six dimensions with an average of 4.19. Faithfulness is the lowest dimension for every backbone and visual specificity the highest.

Subjective evaluation. Table 3 reports pairwise human preferences against four baselines. MVAgent's preference rate exceeds 50% on every criterion in each comparison, with averages from 60.06% against ViMax to 80.12% against StoryDiffusion. Against ViMax the rates are 61.39% for cross-scene consistency, 63.02% for semantic following and 55.78% for aesthetic quality, so the larger margins concern consistency and semantic following.

Table 3. Pairwise human preference rates (%) for MVAgent against each baseline on ViMax-Bench.

MVAgent vs.	CSC	SF	AQ	Avg.
Veo 3.1	78.85%	78.38%	71.06%	76.07%
StoryDiffusion	82.79%	79.62%	77.94%	80.12%
HoloCine	65.31%	68.11%	57.45%	63.49%
ViMax	61.39%	63.02%	55.78%	60.06%

Table 4. Ablation on ViMax-Bench with consistency as in Table 1 and preference rates (%) of the full model against each variant under the protocol of Table 3.

Variant	Consistency			Subjective evaluation (%)			
	CC	IC	GC	CSC	SF	AQ	Avg.
Portraits only	0.5286	0.5597	0.5360	71.5	56.3	53.9	60.6
w/o multi-view library	0.5319	0.5624	0.5388	69.1	55.8	52.7	59.2
Planned state only	0.5382	0.5690	0.5417	67.2	54.3	52.9	58.1
Ours (w/o RL)	0.5239	0.5496	0.5204	63.6	54.4	51.1	56.3
Episode-level RL	0.5362	0.5681	0.5403	55.0	52.9	51.2	53.0
Ours	0.5417	0.5712	0.5438	-	-	-	-

Ablation. Table 4 evaluates conditioning components and training configurations. The conditioning variants use portraits alone, retain the environment image without the multi-view library, or replace observed character states with planned states. The full model achieves the highest CC, IC and GC and is preferred over every variant, from 60.6% against portraits alone to 58.1% against planned states, so the margin shrinks as components are added. The training comparison contrasts the untrained base policy, episode-level RL, which groups complete-video rollouts under a terminal reward per video, and Trunk-GDPO, which forms candidate groups at individual shot states. The full model is preferred over the untrained policy in 56.3% and over episode-level RL in 53.0% of comparisons, chiefly on cross-scene consistency (63.6% and 55.0%). Within each group, both metrics agree on the ranking.

4. CONCLUSION

MVAgent formulates multi-shot narrative video generation as condition construction for a frozen generator, carried out by agents that exchange typed conditioning inputs. A Spatial Grounding agent supplies reusable spatial references, an Observer and a Transition agent turn VLM observations into continuity memory and action and position references, and Trunk-GDPO trains the Orchestrator that composes them into requests with shot-level feedback, subject to the generator's capabilities and the judges' reliability.

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